

Multi-Tier Computing: Front-End Tools, E-Commerce & Enterprise Architectures

CO5 Discussion Points – Exam Notes

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Front-End Tools

Essential / Expected Features

1. Integrated Development Environment (IDE)

An IDE used in multi-tier front-end development must support:

- a. **Designing** (preferred)
- b. **Programming**
- c. **Debugging**
- d. **Testing**
- e. **Global Project Management features** for global multi-teams

2. Development Framework Integration

A front-end tool should integrate with development frameworks appropriate to the application domain (e.g., e-commerce platforms, enterprise portals).

E-Commerce Front-End Requirements

Navigation and Hierarchy

1. **Three-tier hierarchy:**
 - a. Category / Subcategory / Product
2. **User-friendly, clear navigation.**
3. **Easy integration** of e-commerce systems and easy to integrate into an existing site.
4. **Ability to have multiple navigation** based on Manufacturers, Categories, and Products.

Home Page Features

- a. **Category selection:** Selection of products based on categories.
- b. **Featured items display:** Display of featured items.
- c. **Sale items display:** Display of sale items.
- d. **Manufacturer listing:** Display of manufacturers.
- e. **Content Management System (CMS):** To add content-managed pages.

Shopping Cart Features

- a. Quantity update
- b. Indication of savings for sale items
- c. Shipping method selection
- d. Redeeming coupons

Checkout Features

- a. Ability to **transfer shipping field data into billing fields**
- b. **Express checkout** for returning customers

- c. **Checkout indicators:** Progress indicators at the bottom of each page throughout checkout

Order Review Page Features

- a. **Easy to edit information:**
 - i. Easy-to-read display of shipping/billing/payment information with ability to edit.
- b. **Easy listing on order review page:**
 - i. Itemised list of products along with tax, shipping, and total.
- c. **Ability to add comments to order:**
 - i. Text area for delivery instructions and other customer comments.
- d. **Assignment of new password to customers:**
 - i. Input of password for new customers.

Sales Force Automation Architecture

Overview

The Sales Force Automation (SFA) architecture focuses on three core qualities:

- **Interoperability**
- **Scalability**
- **Management of tools and software**

Each of these is supported by **3-tier / n-tier architecture capabilities**.

Key Architectural Contexts Illustrated

The document illustrates several real-world enterprise architecture examples using n-tier and multi-tier principles:

1. **Sales Force Automation Architecture:** An architecture that demonstrates interoperability, scalability, and tool management through layered n-tier capabilities.
2. **N-tier Application for Sales:** Architecture of an n-tier application designed for sales persons in a company, supporting distributed sales operations across tiers.
3. **Accounting Software Architecture using SharePoint:** A multi-tier architecture implementing accounting software on a SharePoint platform, separating presentation, business logic, and data concerns.
4. **Courseware Architecture:** An n-tier architecture applied to courseware delivery systems, enabling distributed content access and management.

Summary of Key Themes

- **Front-end tools** for multi-tier systems must support the full development lifecycle: design, programming, debugging, testing, and global project management.

- **E-commerce systems** require careful front-end design across navigation, home page, shopping cart, checkout, and order review — each mapping to distinct responsibilities in the presentation tier.
- **Enterprise architectures** (Sales Force, Accounting, Courseware) all leverage 3-tier/n-tier models to achieve interoperability, scalability, and maintainability.
- The **3-tier hierarchy** (Category/Subcategory/Product) in e-commerce directly mirrors the logical tier separation concept in multi-tier computing.